

INCLUSIVE EMPLOYMENT PLATFORM

Accessible, AI-Assisted Recruitment for Persons with Physical Disabilities

Final Engineering Presentation

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Capability and opportunity remain disconnected

Digital recruitment can reproduce exclusion when accessibility, trust and task-level ability are missing.

1.3B

people experience significant disability worldwide

30% lower

labour-market participation than persons without disabilities

>80%

of 700+ surveyed candidates named identifying inclusive employers as the greatest barrier

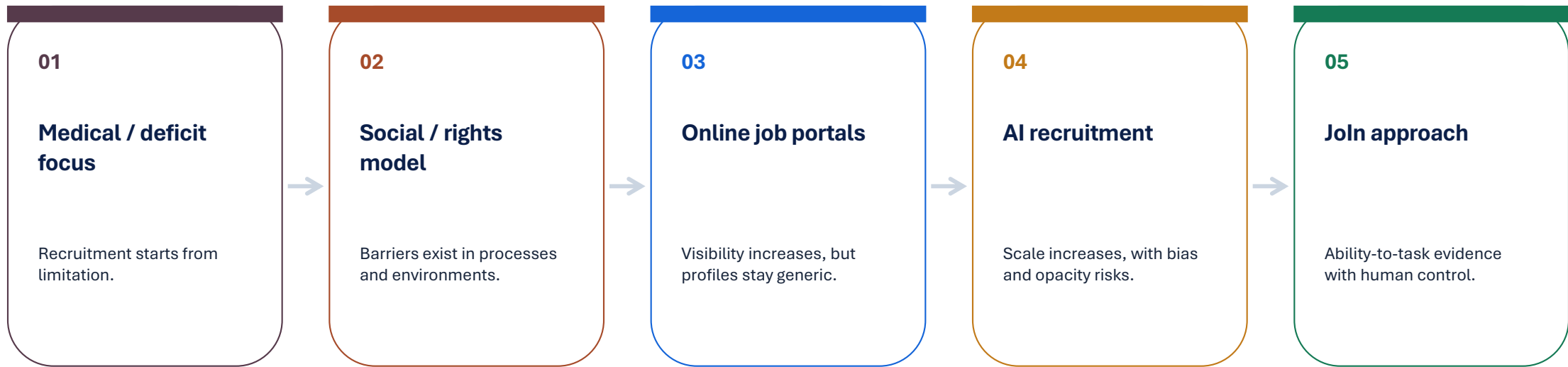
The engineering problem is not a lack of ability.

It is the distance between candidates, accessible recruitment processes, structured job tasks, workplace support and employers prepared to evaluate ability.

Sources: WHO Disability and Health (1.3B / 16%); ILO Global Disability Summit 2025 (30% participation gap); Business Disability Forum / Evenbreak survey (>700 respondents).

From disability-as-deficit to accessible decision support

The project follows the social and rights-based view: barriers can be changed through work design and technology.



Software contribution: change the recruitment interaction and explanation, not the candidate.

Related approaches and the remaining gap

A synthesis of solution categories studied in Phase I and Phase II; not an exhaustive commercial benchmark.

General job platforms

Strength: reach and familiar workflows.
Gap: generic profiles and limited disability-specific support.

Inclusive job portals

Strength: targeted visibility and employer awareness.
Gap: listings may not explain task-level feasibility.

Automated AI screening

Strength: speed and scale.
Gap: data quality, bias, privacy and individual explainability.

Human coaching

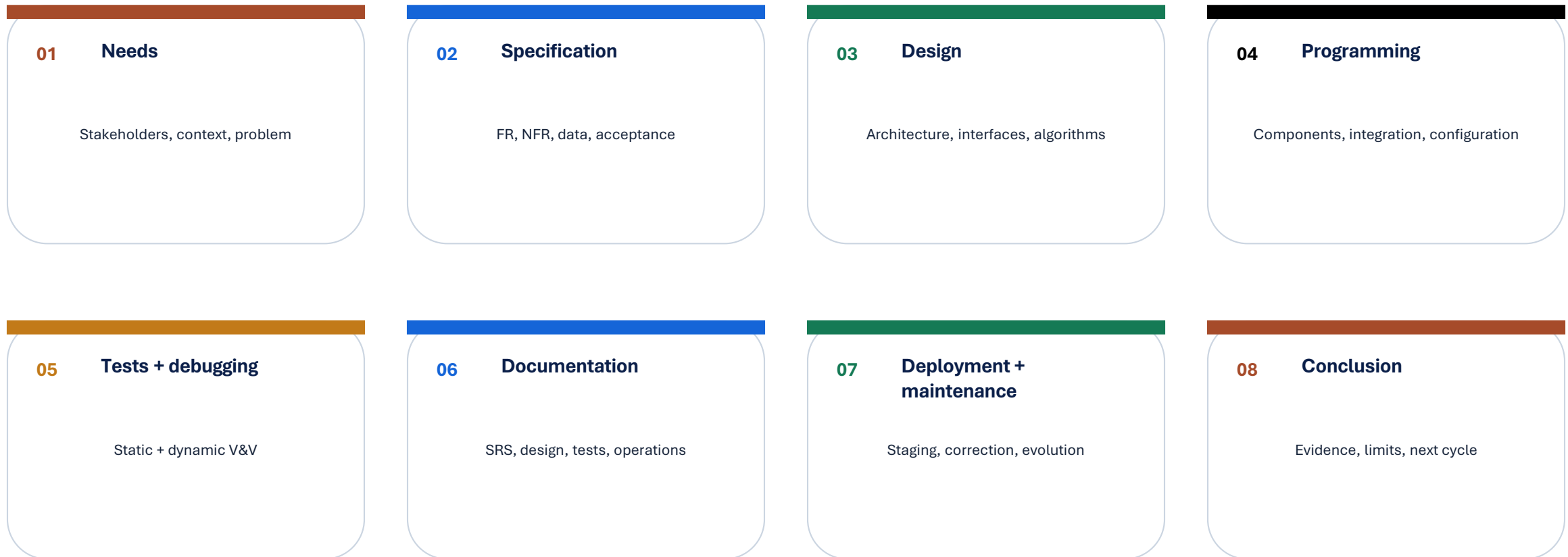
Strength: contextual judgement and adaptation.
Gap: intensive case-by-case work is difficult to scale digitally.

Research gap

One accessible workflow joining controlled hospitality tasks, candidate abilities, workplace assistance, explainable matching and human review.

An iterative lifecycle with V-model traceability

The lifecycle is evolutionary; verification and validation are prepared from the specification stage and repeated after material change.



Chosen process: evolutionary prototyping + iterative increments | Quality control: requirement-to-test traceability inspired by the V-model

Needs came from documents, prototypes and real users

The target need was refined from multiple sources; stakeholder feedback did not replace standards or measurable engineering evidence.

People and stakeholders

- Candidates with varied access needs
- Employers and hospitality teams
- Verifier and administrator roles
- Supervisors and institutional context

Documents and domain sources

- Phase I artefacts and feedback
- WCAG 2.1 and accessibility research
- Hospitality job descriptions and tasks
- Security, privacy and deployment constraints

Elicitation techniques

- Background reading
- Iterative prototyping and review
- Scenario-based task observation
- Arc en Ciel and JAWS evaluation



Needs that shaped the product

- Low literacy and digital confidence -> simple flows and voice support
- Motor or visual barriers -> keyboard, reflow and assistive-technology support
- Privacy and trust -> consent, private evidence and human confirmation
- Employer uncertainty -> controlled tasks and explainable compatibility

What Phase II delivered

The final prototype expands Phase I into a multilingual, multimodal and cloud-deployed engineering system.

4

authorized user roles

15

hospitality positions

923

catalogue tasks

14,453

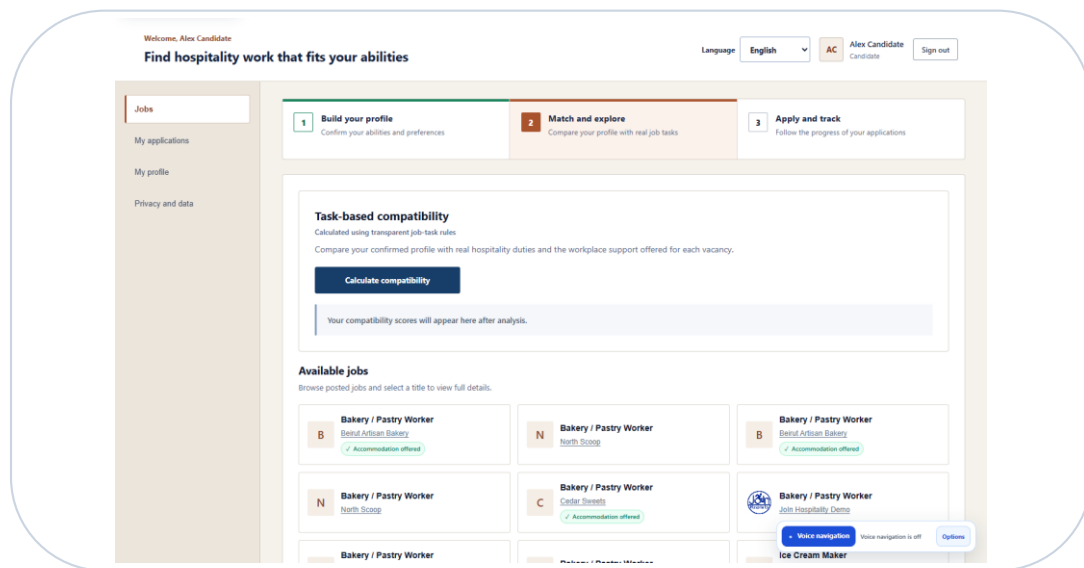
task assessments

74

published opportunities

3

interface languages



Integrated outcome

Candidate-controlled AI profile assistance
 Jobs, hospitality training, or both
 Deterministic explainable compatibility
 Employer requirements and assistance
 Verifier and administrator oversight
 English, French and Arabic with RTL

Requirements were measurable, reviewable and traceable

The SRS states what the system must do and how well it must do it, without prematurely prescribing implementation.

SRS quality criteria used

- Unambiguous
- Complete
- Consistent
- Verifiable
- Modifiable
- Traceable
- Maintainable

NEED / REQUIREMENT	DESIGN + IMPLEMENTATION	TEST / EVIDENCE
Accessible candidate workflow	Semantic React controls + translated UI	18/18 workflows + JAWS
Role isolation and private evidence	Symfony JWT/RBAC + private storage	51/51 route probes; 19/20 security
Explainable compatibility	Versioned deterministic scoring policy	45/45 staging decisions
Multilingual human-controlled AI	Consent + schema validation + review	60 cases; 53 pass; 7 retained defects

Traceability chain: stakeholder need -> specification -> design decision -> code/service -> test oracle -> retained evidence

Functional requirements implemented by role

The platform is a coordinated workflow, not a collection of disconnected pages.

Candidate

- Register and verify email
- Submit private evidence
- Build or reset profile
- Choose work, training or both
- Match, explore and apply

Employer

- Manage company profile
- Publish job or training offer
- Set education and abilities
- Prioritize tasks and assistance
- Review and update applications

Verifier

- Access authorized requests
- Retrieve private evidence
- Approve or reject verification
- Record auditable decisions

Administrator

- Manage users and catalogue
- Import validated role datasets
- Inspect jobs and applications
- Maintain controlled definitions

Non-functional requirements and measurable qualities

Quality attributes were connected to controls and evidence rather than treated as general promises.

Accessibility

- WCAG 2.1 AA target
- Keyboard and assistive technology
- Responsive reflow and reduced motion
- Clear status and error recovery

Security + privacy

- JWT and server-side RBAC
- Private documents and bounded uploads
- Consent, export, withdrawal and erasure
- Audit events without sensitive payloads

Reliability + performance

- Health endpoints and Docker boundaries
- Repeatable p50 / p95 measurements
- No silent rollback on email failure
- Transparent cold-start and latency limits

Maintainability + i18n

- Component-based React interface
- Controller-service-repository backend
- Versioned scoring policy
- Three locales with stable codes and IDs

WCAG 2.1 Level AA: target, implementation and evidence

WCAG organizes success criteria under four principles; Level AA includes all applicable Level A and AA criteria.

Perceivable

- Labels and alternatives
- Structure and headings
- Status not based on colour alone
- Contrast target

Operable

- Keyboard workflows
- Visible focus
- No pointer-only critical path
- Reduced-motion support

Understandable

- Consistent navigation
- Instructions and linked errors
- Confirmation and recovery
- Plain-language feedback

Robust

- Semantic controls
- ARIA status / alert patterns
- Stable focus management
- JAWS-tested workflow

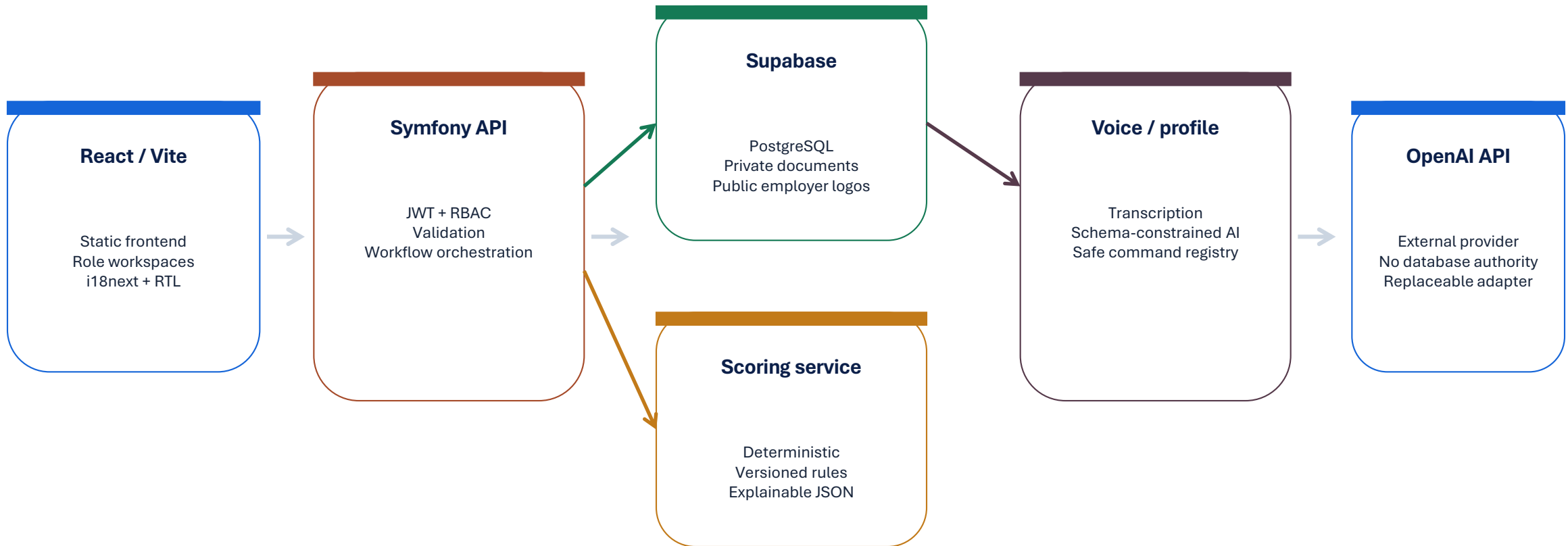
Evidence

18/18 accessibility workflow cases completed | 104 automated route-layout scenarios | JAWS registration and application completed

Honest boundary: 16 scenarios retained axe findings and 2 high-zoom scenarios showed overflow; therefore this is a WCAG 2.1 AA target, not a full conformance claim.

Cloud architecture and explicit trust boundaries

The browser presents the interface; Symfony remains the authoritative security and business-workflow boundary.



Shared service tokens protect Symfony-to-Python calls; secrets stay server-side and HTTPS protects public origins.

Design patterns used to control complexity

The implementation combines patterns at the presentation, application, integration and decision layers.

Layered / MVC

React presentation, Symfony APIs, Doctrine persistence, isolated Python services.

Controller-Service-Repository

HTTP, orchestration and data access remain separately testable.

Adapter / Gateway

OpenAI is accessed through bounded replaceable service contracts.

Strategy / Policy

Versioned weights, assistance factors and eligibility gates.

Command Registry + FSM

Allowlisted voice actions and mutually exclusive interaction states.

Audit + Immutable Snapshot

Consent, verification, status events and score context remain traceable.

Engineering value: lower coupling, safer external AI, reproducible decisions and independent verification.

Sensitive data is controlled by the backend, not the interface

Frontend checks improve usability; Symfony authorization remains decisive for every protected request.

Identity and role isolation

Email verification
JWT via X-Auth-Token
Candidate | Employer | Verifier | Admin
Anonymous, forged and cross-role rejection

Private evidence and uploads

Private candidate-verification bucket
Private application documents
Public employer logos only
MIME and size restrictions

Consent and auditability

Explicit AI consent
No unconfirmed suggestion saved
Export, withdrawal and account erasure
Audited status and verifier decisions

Evidence: 51/51 prerequisite route checks passed | Final security register: 19/20 | Open upload-validation defect retained

Implementation separated concerns and protected behaviour

Coding was one lifecycle activity: integration, configuration management, refactoring and repeatable tests were developed with it.

Implemented components

- React/Vite presentation layer
- Symfony/Doctrine business boundary
- Python deterministic scoring
- Python voice/profile assistance
- PostgreSQL and object storage

Configuration and integration

- Version-controlled staging branch
- Docker service boundaries
- render.yaml infrastructure as code
- Environment-only secrets and JWT keys
- Versioned migrations and datasets

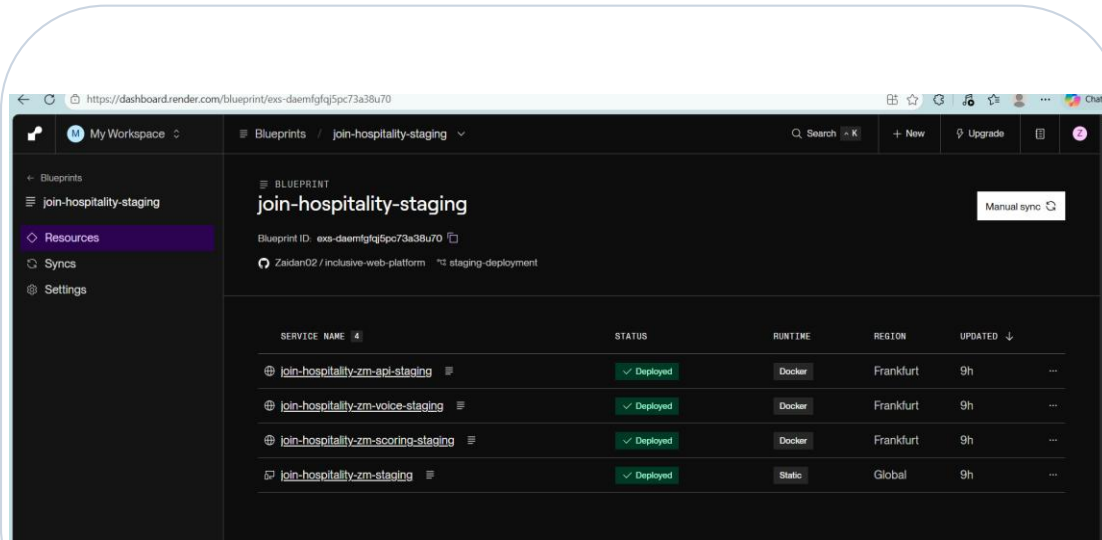
Refactoring discipline

- External behaviour preserved
- Smaller cohesive services
- Adapters localise OpenAI changes
- Command registry limits voice actions
- Regression tests before and after fixes

Design objective: low coupling, high cohesion and localised change - not simply more source code.

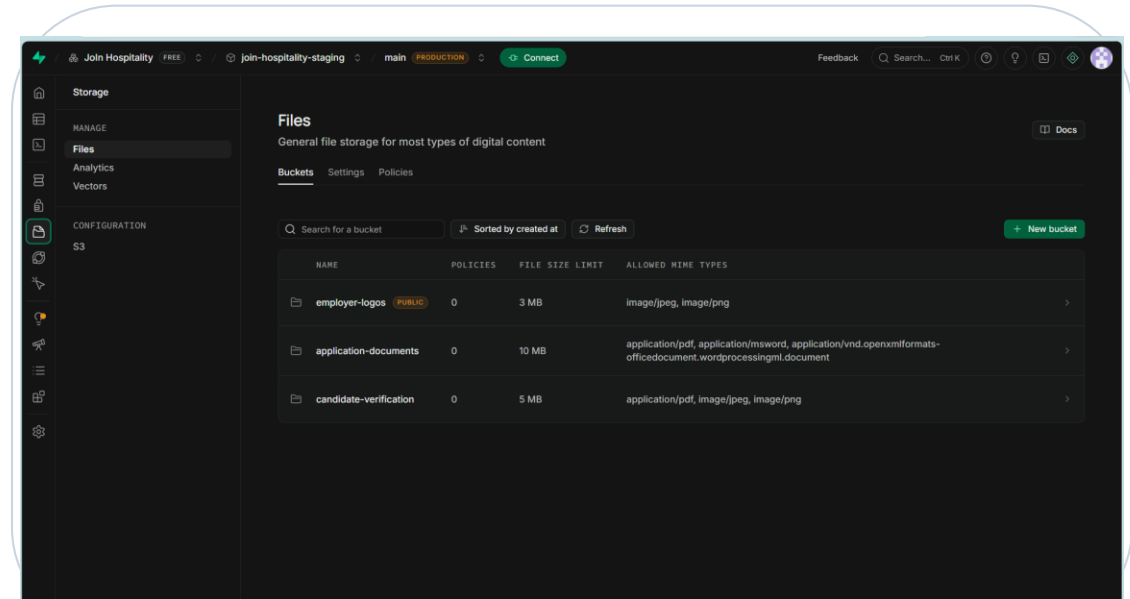
Render and Supabase staging topology

The deployed system proves multi-service integration over public HTTPS; it is a staging prototype, not a production SLA.



Render Blueprint

React static site + Symfony API + voice/profile service + scoring service

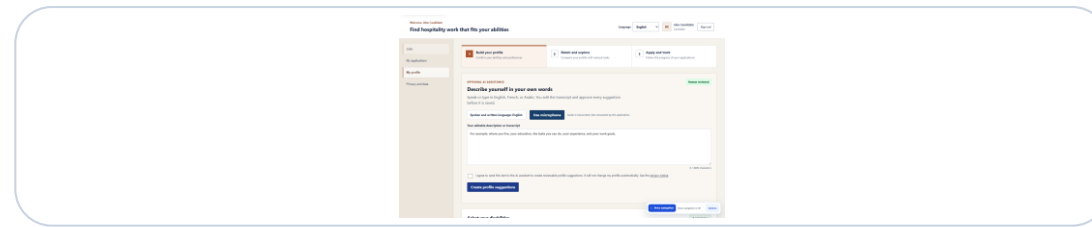
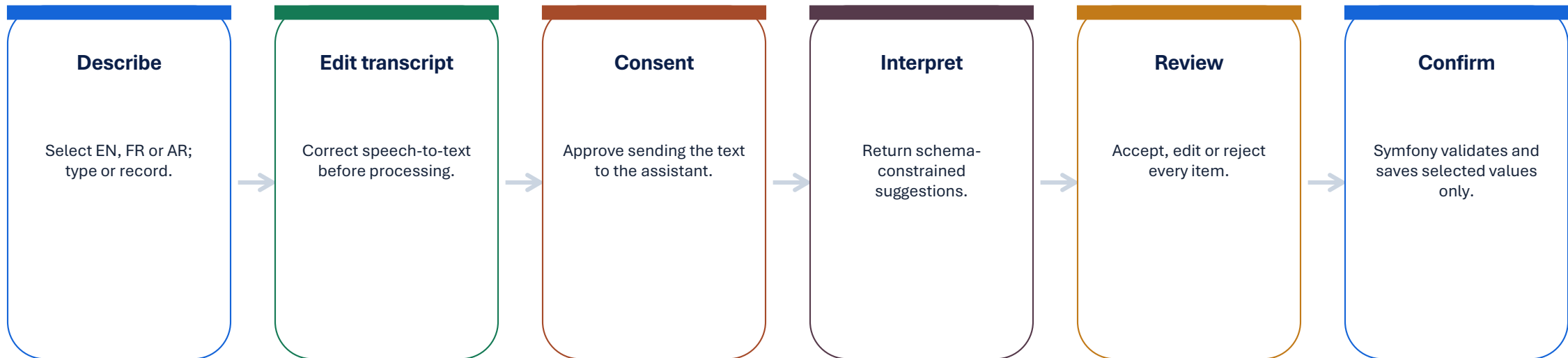


Supabase data layer

SSL PostgreSQL pooler + three purpose-specific storage buckets

AI-assisted profile: advisory, consent-gated and reversible

The candidate can type or speak; every AI result remains a suggestion until explicit confirmation.



Structured suggestions in English, French and Arabic

Sixty gold-standard narratives covered clean and imperfect wording, abilities, preferences, positions, tasks and safety boundaries.

OPTIONAL AI ASSISTANCE

Human reviewed

Describe yourself in your own words

Speak or type in English, French, or Arabic. You edit the transcript and approve every suggestion before it is saved.

AI profile input language

English

This may be different from the interface language.

Use microphone

Audio is transcribed, then discarded by this application.

Your editable description or transcript

My first name is Maya and my last name is Haddad. My phone number is 70123456 and I live in Beirut. Set my About you text to: I am reliable, patient, and enjoy helping guests. I completed high school. I can read, write, and count independently. I want both paid work and a hospitality training program. I am interested in Cashier and know that position independently. I explicitly have a Hand disability. I can prepare the cash system and box, check and test electronics, and identify Food and Beverage menu components.

519 / 4000 characters

☒ I agree to send this text to the AI assistant to create reviewable profile suggestions. It will not change my profile automatically. See the [privacy notice](#).

Create profile suggestions

ENGLISH

ASSISTANCE IA FACULTATIVE

Validation humaine

Décrivez-vous avec vos propres mots

Parlez ou écrivez en anglais, en français ou en arabe. Vous pouvez modifier la transcription et approuver chaque suggestion avant son enregistrement.

Langue de saisie du profil IA

Français

Elle peut être différente de la langue de l'interface.

Utiliser le microphone

L'audio est transcrit, puis supprimé par cette application.

Votre description ou transcription modifiable

Mon prenom est Maya et mon nom est Haddad. Mon numero est 70123456 et j'habite a Beyrouth. Pour mon texte A propos: Je suis fiable, patiente et j'aime aider les clients. J'ai termine le lycee. Je sais lire, ecrire et compter de facon autonome. Mon choix explicite est les deux: un emploi remunere et une formation en hotellerie. Je souhaite travailler comme caissiere et je connais ce poste de facon autonome. Je declare explicitement un handicap Hand. Je sais preparer le systeme et la caisse, verifier et tester les appareils electroniques, et identifier les composants du menu Food and Beverage.

598 / 4 000 caractères

☒ J'accepte d'envoyer ce texte à l'assistant IA afin de créer des suggestions de profil à vérifier. Mon profil ne sera pas modifié automatiquement. Consultez l'[avis de confidentialité](#).

FRENCH

مساعدته اختيارية بالذكاء الاصطناعي

خاضع لمراجعة بشرية

عرّف عن نفسك بكلماتك الخاصة

تحدث أو اكتب بالإنجليزية أو الفرنسية أو العربية. يمكنك تعديل النص والموافقة على كل اقتراح قبل حفظه.

لغة إدخال ملف الذكاء الاصطناعي

العربية

يمكن أن تختلف عن لغة الواجهة.

استخدام الميكروفون

يُحوّل الصوت إلى نص، ثم تختص فيه هذه الميزة.

وصفك أو النص القابل للتعديل

اسمي الأول مايا واسم العائلة حداد. رقم هاتفي 70123456 وأسكن في بيروت. اكتب في نبذة عني: أنا موثوقة وصبورة وأحب مساعدة الضيوف. أنهيت الثانوية. أستطيع القراءة والكتابة والعقد باستقلالية. أريد عملاً مدفوعاً وبرنامج تدريب في الضيافة، أي الاثنين. أنا مهتمة بوظيفة كاشير وأعرف هذه الوظيفة باستقلالية. أصرح أن لدي إعاقة Hand في اليد. أستطيع تجهيز نظام وصندوق النقد، وفحص الأجهزة الإلكترونية واختبارها، ومعرفة مكونات قائمة الطعام والشرب.

4000 / 427 حرف

☒ أوافق على إرسال هذا النص إلى مساعد الذكاء الاصطناعي لإنشاء اقتراحات ملف قابلة للمراجعة. لن يُعدّل ملفي تلقائياً. راجع [إشعار الخصوصية](#).

إنشاء اقتراحات للملف

ARABIC + RTL

53/60 passed (88.3%) | 7 failures retained | Main weakness: explicit Both work-and-training preference could be omitted

Voice command language

High confidence · Evidence: States that her first name is Maya.

AI-generated voice

Langue des commandes vocales

Maya

Voix générée par l'IA

Confiance élevée · Justification : Elle indique que son prénom est Maya.

لغة الأوامر الصوتية

Maya

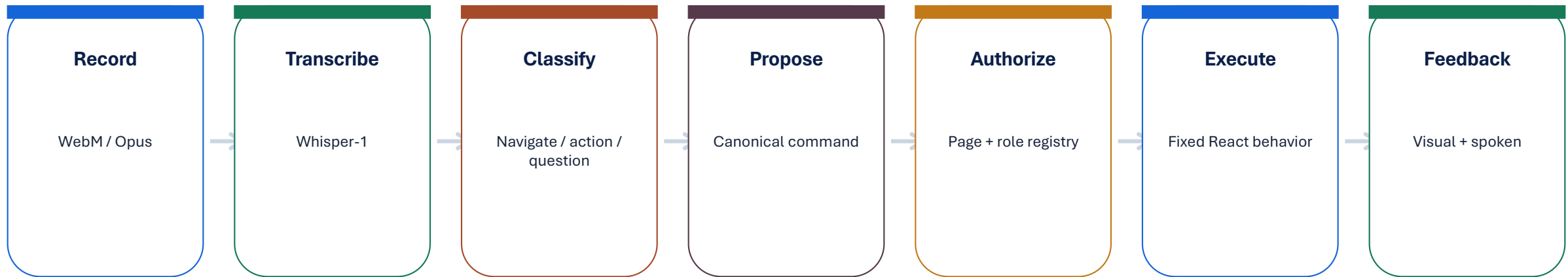
صوت مولّد بالذكاء الاصطناعي

إضافة اسم العائلة

تعديل القيمة المعقّرة

Bounded voice-navigation execution path

Natural language proposes an action; only the deterministic registry may authorize and execute it.



Safety boundary

Unregistered routes and cross-role actions are rejected. Submission and reset require confirmation. Context is sanitized and bounded to six turns.

Evidence: 57/57 automated registry/action-safety checks passed. Quiet commands worked reliably; background noise and dictated email characters required correction in some trials.

Deterministic compatibility: explainable by construction

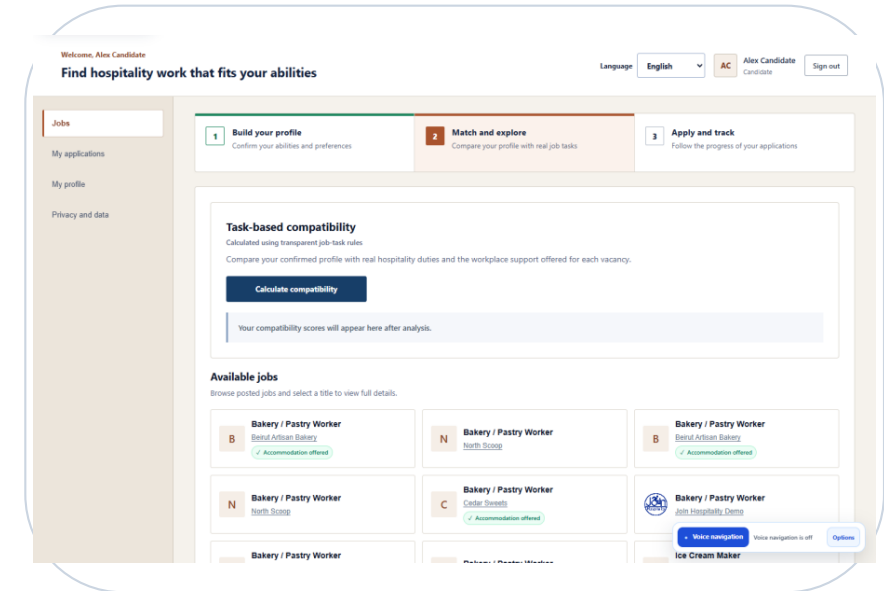
Matching uses controlled expert assessments rather than a learned model trained on unavailable employment outcomes.

Task score

$\text{Sum}(\text{adjusted weight} \times \text{feasibility factor})$

$\text{Sum}(\text{adjusted weight}) \times 100$

Feasibility: 1.00 feasible | 0.75 with offered assistance | 0.00 avoid
 Important task multiplier: 1.5
 Practical abilities: 25% and education: 10% when configured
 Unsafe mandatory tasks and required abilities remain explicit gates

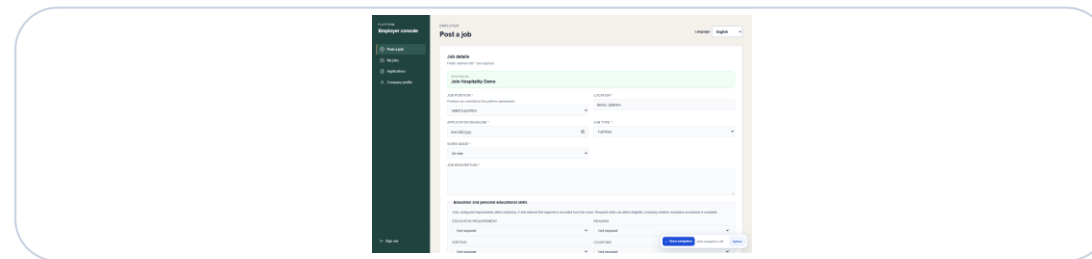
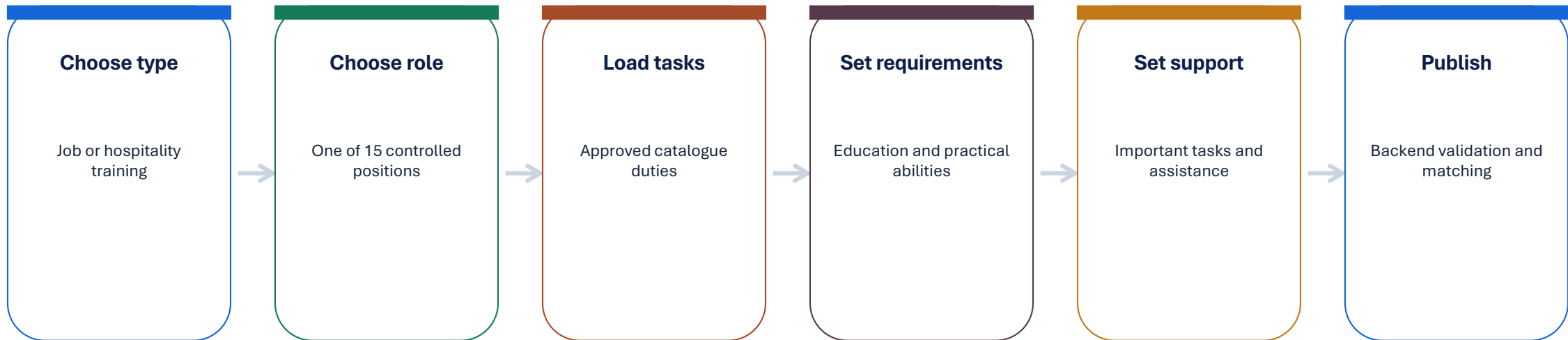


45 / 45

staging formula, eligibility and determinism cases passed

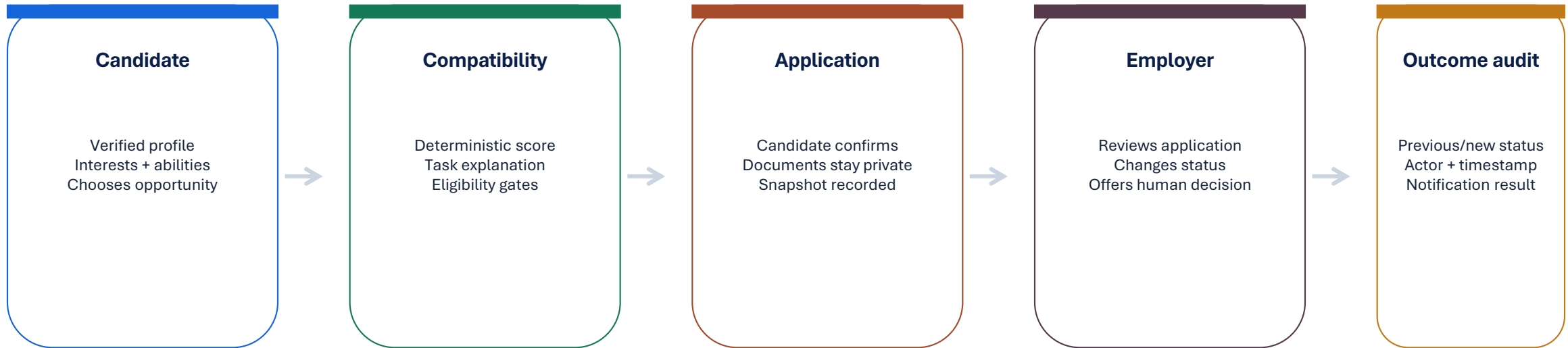
From controlled role to published job or training program

Employers configure the real opportunity; the catalogue supplies consistent tasks and matching inputs.



Human-controlled recruitment and auditable outcomes

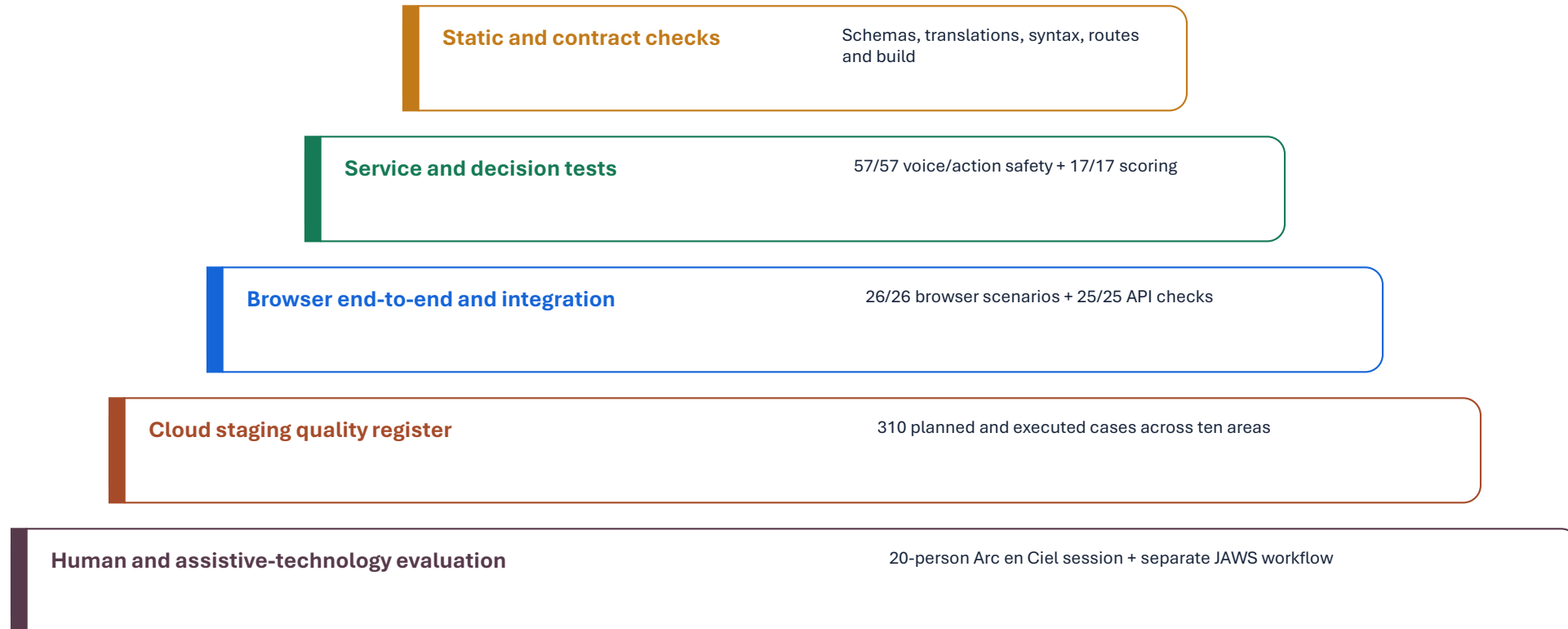
Each role contributes a bounded decision; no single AI response creates an employment outcome.



Verifier authorizes disability-card evidence | Administrator maintains controlled catalogues and users

Verification and validation across the lifecycle

Verification asks whether specifications were implemented correctly; validation asks whether the right system was built for users and context.



All 310 planned staging cases were completed

The register preserves first-run failures and noisy partials instead of hiding them behind the final result.

93.2%

FINAL PASS RATE

289 final passes
21 retained failures
0 blocked / not run

First execution

274

passed directly

15

partial in noisy transcription

21

failed and retained

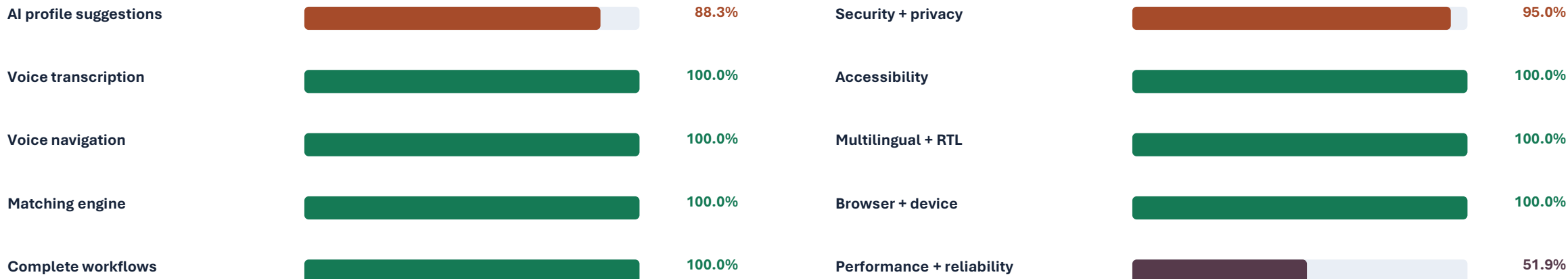
Retest interpretation

The 15 noisy cases passed after repetition or manual correction. Failures were not silently reclassified; they remain visible in the workbook and report.

Evidence: Join Final Test Register - Master Register, Dashboard, Defects and Evidence Index.

Results by engineering test area

Final pass rate is useful only when read together with each area, target and open defect.



Read the pattern, not only the average

Strong functional correctness and language coverage. Improvement work is concentrated in AI preference extraction, one upload behavior, and staging latency under load.

Testing produced actionable defects, not only pass counts

A final-year engineering project should explain what failed, why it matters and what should change next.

AI preference extraction

Explicit Both work-and-training wording was intermittently omitted.

Contrast and zoom reflow

Helper text contrast and high-zoom candidate header overflow need remediation.

Oversized upload behavior

PHP could drop the file while an application record was still created.

Write-operation latency

Publishing and application p95 exceeded the one-second target in four cases.

Free-tier cold starts

Healthy services may require warm-up; temporary 503s were observed.

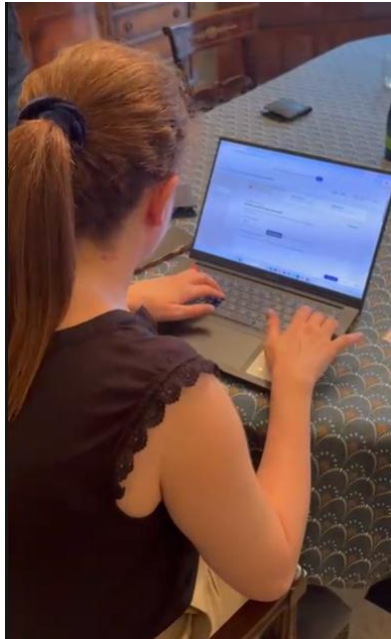
Acoustic robustness

Noise and dictated email characters can require repetition or editing.

Engineering response: preserve failures, prioritize by risk, correct, retest and retain traceable evidence.

The platform was evaluated with its intended users

Human evidence complements automated checks; photographs document sessions but do not prove universal accessibility.



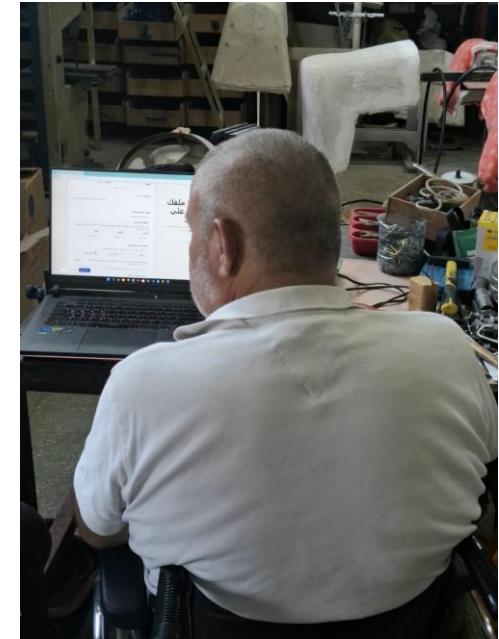
JAWS evaluation

Noelle completed navigation, registration and job-application interaction on her adapted device.



Arc en Ciel: 20 participants

Users with different needs interacted with the main platform workflows in a real setting.



Accessibility in context

Simple steps, Arabic support and voice assistance helped broaden participation.

Functionality held; free-tier latency did not always meet the target

Availability and response-time quality were evaluated separately.

Functional reliability

144 / 144

shared-load responses succeeded
with zero 5xx

96 / 96

publish/application mutations
returned HTTP 201

The deployed services produced the expected functional output after health warm-up.

Latency and dependency limits

3 / 9

shared-load latency cases met
threshold

4 / 6

write p95 checks exceeded one
second

Free-service cold starts and external AI availability remain staging constraints. Local bundle budgets passed after removing a blocking external font request.

Conclusion: functionally usable staging evidence, but no production availability or capacity claim.

Cost is effort plus infrastructure, external services and change

The prototype used free staging resources; production planning must estimate both fixed engineering effort and variable operational demand.

Estimation approach

- Bottom-up by subsystem and lifecycle activity
- Cross-check with analogous deployments
- Use scenarios for active users and AI calls
- No unsupported COCOMO claim without calibrated effort history

Main cost drivers

- Human engineering and verification effort
- Non-sleeping compute and managed database
- Storage, email and monitoring
- OpenAI transcription and structured suggestions
- Accessibility and security maintenance

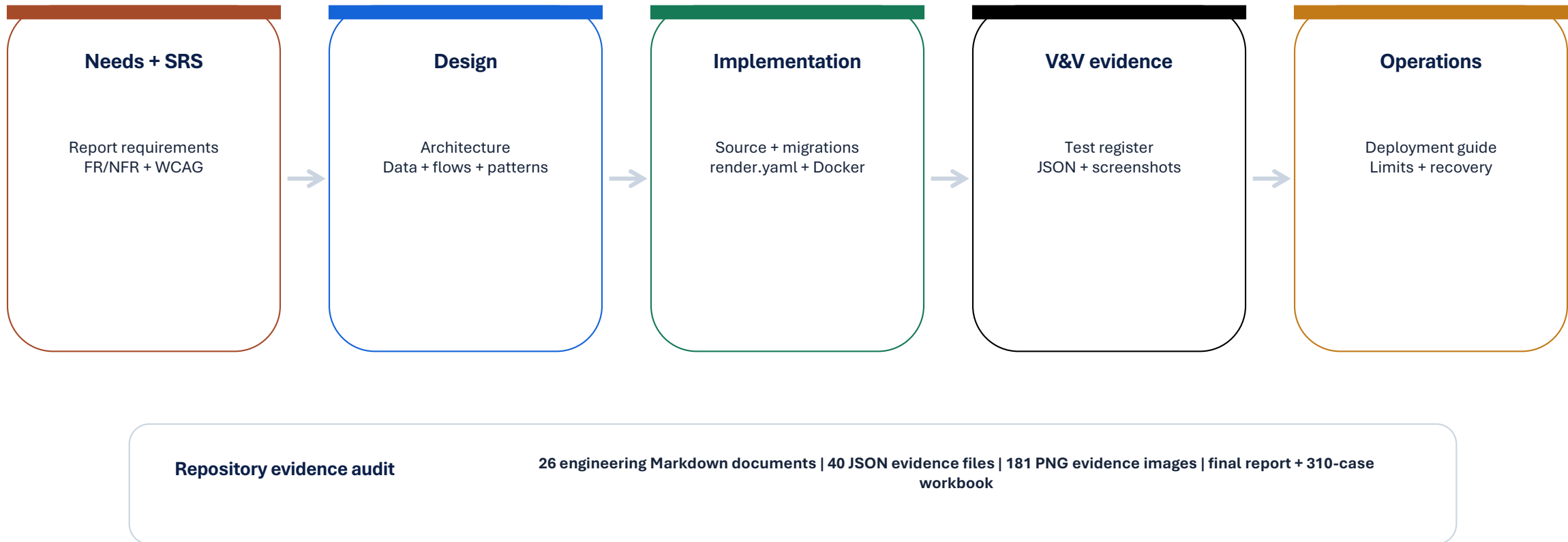
Controls and trade-offs

- Per-user daily AI and voice quotas
- Rate limits, cooldowns and budget alerts
- Lower-cost model routing and manual fallback
- Measure queries before caching
- Cache only repeated non-sensitive reads

Maintenance categories: corrective defects | adaptive provider/platform change | perfective features | preventive refactoring and monitoring

Every lifecycle stage produced a reviewable artefact

Documentation supports communication, verification, change control, deployment and future maintenance.



A validated bridge from ability to opportunity

The project demonstrates technical feasibility and practical relevance while keeping its limitations explicit.

Delivered

- + Four protected role workflows
- + Multilingual and RTL interface
- + Candidate-controlled AI and voice assistance
- + Deterministic task-level matching
- + Jobs and hospitality training programs
- + Render/Supabase cloud staging
- + 310-case register plus human validation

5-minute demonstration

- 00:00 Candidate profile + consent
- 01:00 Match score + explanation
- 02:00 Employer job/training workflow
- 03:00 Verifier/admin + deployment evidence
- 04:00 Language, keyboard and voice path

If the paid AI dependency is unavailable, use the retained multilingual screenshots and continue the deterministic live workflows.



Thank you

Questions?

<https://join-hospitality-zm-staging.onrender.com/>

Accessible recruitment | Explainable matching | Human control